

AI x Crypto ~ Active Entity Ontology for Science & Active Blockference ~ A

Active Entity Ontology for Science & Active Blockference ~ Active Inference Institute | 15:55

Hi everyone, I'm Jakub and today I'm going to give a very brief overview of some projects that we've been developing at the Active Inference Institute and their relevance to decentralized AI. Now, I think my talk and Alex's right after me is going to overlap slightly. So, I'm going to try to keep the theory of Active Inference and one particular project to a minimum, although I'll try to give some brief introduction to what Active Inference is so that you have the relevant background. So, since this is a mini conference on decentralized AI, I'm going to assume that most of you have at least heard of reinforcement learning, which on the computational side is quite similar to Active Inference, but we can think of Active Inference as being a kind of generalization of reinforcement learning where agents interact with their environment and try to solve a certain task or try to navigate in their environment. But instead of maximizing some utility function or instead of their goal being always maximizing for reward, Active Inference agents are perhaps more general where the kind of utility function or the general function of Active Inference agents is always uncertainty minimization. So, Active Inference agents form some kind of internal model of their environment and they can learn more about their environment by engaging with it through action and then observing what kind of sensory input goes from the environment to the agents. So, there's a kind of duality between perception and action that's always mediated through uncertainty minimization. And because of its generality, this framework can be applied to pretty much any system that's salient enough to persist through time, which was kind of the motivation for the first project, which Alex will talk about a bit more, so I'm only going to mention it briefly, which is the Active Entity Ontology for Science, which really stands at the intersection of three fields, namely DSI Systems Engineering and Active Inference, where our aim was to create an ontology that individuals and communities within DSI can use to collectively minimize their uncertainty about what kind of system they find themselves in and how that system might evolve through time. And I think this is the same can be done for decentralized AI, which has a lot of similar artifacts that have to do with Web3 and perhaps we can say that either DSI is encapsulated within decentralized AI or the other way around, depending on which you think is the bigger set. But the core utility of having a shared ontology that's kind of grounded in another formal framework, namely active inference, allows us to more specifically define what kind of entities and artifacts we can find in a system and how these entities interact with each other and minimize their uncertainty. So we can kind of have a better understanding of the dynamics of our system and be able to better make decisions about what kind of protocols we should create or how should our community interact with other communities and so on. So this also leads me to the active block friends project which aims, which is kind of on the computational side of AOS where we aim to combine the open source framework CAD CAD which was developed by block science and that's been used mainly for token engineering and also just generalized dynamical systems

[illegible]

So that would be a very brief overview of some of the projects that we're developing at the Active Inference Institute and if you have any questions then I'm not sure whether we have the time but thanks a lot for your attention.

Okay I'm not seeing any text questions yet so I'll definitely invite people to ask you any questions that they have about that either in the 10-minute session or towards the end when we finish.

Very cool to hear what you're doing. I'll bring up your colleague Alex Rietken now.

Yeah fine thanks.

Cool take it away.

Can you see it right?

Okay so I will add here some information about Active Inference and AEOs and continue to talk to Jakob talk but also we'll provide more information from Institute Scale about our work and possible collaboration with our participants. So we are Institute which is working on the frontier of

[illegible]

we as an agent to provide more interfaces and more structured ways to communicate in this frontier domains because it's quite hard science and we need to find ways to use common language in different projects and by that evolve evolve research ecosystems

so i think that's all from from from my side so if you have some questions

can you hear me

[illegible]

Cool, thank you.